Does Quantum Mutual Information Predict the Quantum–Classical Performance Gap in Spin-Network Reservoir Computers?

A Task-Agnostic Diagnostic Tested Across Decoherence Channels on the NARMA-10 Benchmark

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Abstract. Quantum reservoir computing (QRC) uses the dynamics of a driven quantum system to process time series, training only a linear readout. Which property of the quantum state makes a reservoir useful is still debated: entanglement, coherence and total correlations have each been linked to performance, but almost always to absolute performance rather than to the margin over a carefully tuned classical baseline. We ask whether the quantum mutual information of the input-driven reservoir state, computed without access to task targets, predicts the NARMA-10 performance gap between small transverse-field Ising reservoirs and capacity-matched, tuned echo state networks, beyond what entanglement negativity and ℓ_1-norm coherence already explain. We compare this relationship under pure dephasing and amplitude damping, include finite-shot noise throughout, and test whether a fitted diagnostic transfers across decoherence channels, system sizes and tasks. [TODO: Summarise the main results and their implications once Sec. V is complete.]

I. INTRODUCTION

Many prediction problems in science and engineering involve time series. Solving them requires a model that remembers past inputs and transforms them nonlinearly. Reservoir computing (RC) meets this need by driving a fixed, high-dimensional dynamical system with the input and training only a linear readout of its state [1, 2, 3]. Because the reservoir is never trained, it need not be a simulated network. Any physical system with fading memory and a suitable nonlinear response can serve, which has made RC a leading framework for computing with unconventional hardware [4, 5].

Quantum systems are an attractive substrate. An N-qubit reservoir evolves in a 2^N-dimensional state space, and its dynamics can generate correlations with no classical counterpart [6, 7]. Quantum reservoir computing (QRC) needs no gradient-based training of the quantum dynamics, and it requires some dissipation to obtain fading memory [6, 8]. Both features suit today's noisy intermediate-scale devices [9, 10]. Since the proposal of Fujii and Nakajima [6], QRC has been studied in spin networks, coupled oscillators, single qudits and superconducting processors. Its performance has been linked to the dynamical regime, the readout observables and the effective dimension of the system [11, 12, 13, 14, 15, 16].

It remains unclear which property of a quantum reservoir is responsible for good performance, and whether that property is quantum at all. Entanglement can support short-term memory [17], but its benefit depends on the input frequency once dissipation is present [18], and memory can peak well below maximal entanglement [19]. In a transverse-field Ising reservoir, information processing capacity (IPC) rose monotonically with ℓ_1-norm coherence, while negativity could fall by more than two-thirds without a significant loss of capacity [20]. In the same system, purely classical correlations sustained capacity in the localised regime. Recent preprints propose further candidates, including magic, non-Markovianity, Krylov observability and a memory–nonlinearity balance [21, 22, 23, 24]. Because these diagnostics are computed from the reservoir state alone, one that reliably predicted performance could guide reservoir design before any task is attempted. Which measure, if any, can play that role is not settled.

Two factors complicate the search for such a diagnostic. The first is the type of decoherence. Non-unital amplitude damping can enlarge processing capacity and improve performance, whereas dephasing and depolarising noise degrade it [14, 25, 26]. Whether the link between a diagnostic and performance also depends on the channel is a separate question. Bit-flip and phase-flip noise produced a single capacity–coherence curve in Ref. [20], but amplitude damping has not, to our knowledge, been compared in this way. The second factor is measurement. Conclusions drawn from exact expectation values can change once finite sampling is included, and omitting shot noise can reverse the apparent effect of decoherence on performance [20, 27, 28].

The choice of benchmark raises a further problem. Claims of quantum advantage have often been made against classical reservoirs tuned less carefully than the quantum model [13, 16, 29]. When echo state networks (ESNs) were matched in feature dimension and tuning budget, the advantage of two QRC mechanisms disappeared in noiseless simulations of up to 11 qubits [30]. A recent survey concludes that no broad advantage over well-matched classical reservoirs has yet been established [31]. For the study of quantumness diagnostics, this means that relating quantum properties to absolute measures, such as memory capacity or IPC, is not sufficient. The quantity of practical interest is the gap between a quantum reservoir and a fair classical baseline, and

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the question is whether a property of the quantum state can indicate in advance when that gap is positive.

We address this question for small transverse-field Ising reservoirs on the NARMA-10 benchmark [32]. We ask whether the quantum mutual information (QMI) of the input-driven reservoir state, computed without access to task targets, explains variation in the performance gap to a matched, tuned ESN beyond what negativity and ℓ_1-norm coherence already explain. We also ask whether this relationship differs between pure dephasing and amplitude damping. QMI is a natural candidate. It is cheap to compute from the state and captures total correlations, both classical and quantum [33]. In Ref. [20] it closely tracked coherence in most regimes, while classical correlations dominated in the localised regime. QMI may therefore carry information that coherence and entanglement miss in some regimes, and be redundant with them in others. Our design is built to tell these outcomes apart.

The study makes four contributions:

1. It tests, using nested cross-validated models and equivalence tests, whether QMI adds predictive information about the quantum–classical performance gap beyond negativity and coherence (H1a).

2. It compares the diagnostic–performance relationship under amplitude damping and pure dephasing, which to our knowledge has not been done before (H1b).

3. It tests whether a diagnostic model fitted in one setting transfers to another channel, a larger system ($N = 8$) and a second task (H1c).

4. It includes finite-shot noise in every primary analysis and uses a classical baseline built to current fair-comparison standards [30, 31]. As a validation step, it first reproduces the IPC–coherence and IPC–negativity results of Ref. [20].

Section II reviews related work. Section III states the hypotheses. Section IV describes the reservoir model, the decoherence channels, the diagnostics, the classical baseline and the statistical analysis. Section V presents the results for each hypothesis, and Sec. VI discusses their implications and limitations.

II. BACKGROUND AND RELATED WORK

A. Reservoir computing and fading memory

Reservoir computing keeps a recurrent dynamical system fixed and trains only a readout, usually by linear regression [1, 31, 34]. The approach needs fading memory: the reservoir state must depend on recent inputs and forget its initial condition, a requirement known as the echo state property [1]. Information processing capacity makes this quantitative [35]. It measures how well the reservoir reproduces orthogonal polynomials of delayed inputs, and its total is bounded by the number of linearly independent output features.

NARMA tasks are a standard benchmark because they need both memory and nonlinearity [32]. NARMA-10, in particular, requires the reservoir to combine inputs from ten steps back. Recent work on a damping-controlled quantum reservoir found that NARMA performance peaked where memory capacity was highest, and proposed memory capacity as a practical indicator of performance [26].

B. Quantum reservoir computing

QRC replaces the classical network with a quantum system. The standard model, introduced by Fujii and Nakajima [6], is a transverse-field Ising network: an input qubit is reset and rewritten at each step, the network evolves, and local observables are read out with time multiplexing. Because unitary evolution is reversible, fading memory requires dissipation; the reset-and-write step supplies it [20]. Dissipation contracts differences between initial states, but its rate matters. Contraction that is too fast erases recent inputs, and contraction that is too slow leaves a dependence on initialisation [31]. Engineered local losses can themselves be a resource, and dissipative spin models form a universal class for fading-memory maps [8].

The nominal state space grows as 2^N, but this overstates what the readout can use: only features exposed by the encoding, dynamics and measurement contribute [31]. Within that limit, the dynamical regime matters. Ergodic dynamics favour tasks needing nonlinear memory, while many-body localisation restricts information flow [11, 20]. Readout design also matters: two-body correlators raise total IPC over single-spin observables [12]. A single d-level system improves as its dimension grows and has outperformed a physically analogous classical reservoir [13]. Reviews [7, 31] list further proposed explanations for QRC performance, from dissipation to chaos and measurement back-action.

C. Quantumness diagnostics and performance

Entanglement. Götting et al. linked entanglement and the occupied phase-space dimension to linear short-term memory. They concluded that high entanglement is a prerequisite for the richer dynamics that make use of the exponential state space [17]. Kora et al. measured entanglement by logarithmic negativity and found that its benefit depends on the input [18]. Without dissipation the entanglement advantage was always present. With dissipation, fast-varying signals gained more from entanglement, and slow signals were better served by a non-entangled reservoir. The benefit also saturated at high entanglement [18]. In a spin network with two input qubits, memory capacity peaked at moderate coupling, well before average entanglement reached its maximum [19].

Coherence. A synthetic reservoir with tunable coherence improved on short-term-memory and parity-check tasks as coherence increased [29]. Palacios et al. found a monotonic relationship between IPC and ℓ_1-norm coherence in the ergodic regime of the transverse-field Ising reservoir [20]. Negativity could fall by more than two-thirds of its noiseless value before capacity dropped significantly. The authors read this as superposition mattering more than entanglement for that reservoir.

Total correlations. Palacios et al. also computed the total

mutual information, the quantum hookup and purely classical correlations [20]. The hookup equals total correlations plus local coherence [36]; local coherence was zero in their system, so the hookup and mutual information coincided. Empirically, both closely tracked ℓ_1 coherence whenever classical correlations were small. Classical correlations carried most of the remaining capacity in the localised regime under strong phase-flip noise. In coupled Kerr oscillators, time-averaged mutual information fell as dissipation rose, while long-term memory capacity improved through more redundant encoding [37]. Correlations and performance can therefore move in opposite directions, depending on the regime.

Other proposed resources. Recent preprints propose further explanations. Dynamics-generated magic is argued to be necessary for computation in qubit-resetting schemes, with the echo state property enforced separately by the Liouvillian spectral gap in Hamiltonian-encoding schemes [22]. Markovian architectures are argued to forget exponentially, and non-Markovian reservoirs are shown to outperform them on tasks mixing short- and long-term correlations [23]. Krylov observability explained Lorenz-63 performance better than IPC when the reservoir was undersampled [24]. Memory restriction, weak measurement, the edge of chaos and dissipation have been unified as tunings of a single memory–nonlinearity balance measured by IPC [21]. Against these, QMI is cheap to compute from the state and is the natural total-correlation quantity, which is why this study focuses on it.

D. Decoherence channels

The evidence favours amplitude damping over unital noise for performance, but mostly on tasks other than NARMA-10. Domingo et al. showed that amplitude damping improved a quantum-chemistry regression task (the first excited-state energy of LiH) at error rates of about 5×10^{-4} or below and shallow depth, while depolarising and phase damping always degraded it [25]. Their explanation is that a non-unital channel adds new Pauli components, whereas unital channels only shrink existing ones. They name time-series forecasting as future work, so this result is not a temporal one.

For temporal tasks, Kubota et al. found that amplitude damping produced time-invariant processing capacity that phase-flip, bit-flip, phase-damping and depolarising noise did not, and that capacity rose with noise level; they confirmed this on IBM hardware [14]. Ricci et al. reported that controllable amplitude damping supports NARMA and Mackey–Glass tasks, while dephasing and depolarising noise progressively degrade memory capacity [26]. The native noise of superconducting processors has also served as the dissipation source for temporal benchmarks [15]. For dephasing, Fujii and Nakajima suggested a constructive role in the reset-and-write map [6, 20], and Götting et al. discuss its effect on performance [17].

Two cautions follow. First, Palacios et al. report that some non-monotonic noise effects found earlier do not survive realistic measurement noise [20]. Second, performance may differ by channel while the diagnostic–performance relationship does not: bit-flip and phase-flip data collapsed onto one capacity–coherence curve [20]. These are separate questions, and H1b addresses the second.

E. Finite sampling

Real reservoirs are read out from a finite number of measurements, and this changes conclusions. Statistical noise can lower total capacity [28, 38]; frameworks for finitely sampled quantum reservoirs are developed in Refs. [39, 40]. Palacios et al. modelled roughly 10^6 shots as Gaussian noise with standard deviation 10^{-3}, and showed that omitting it can reverse the apparent effect of decoherence strength on performance [20]. Kora and Simon found that statistical noise harms performance overall but that entangled, coherent reservoirs are more robust to it than unentangled, incoherent ones, so the benefit of quantumness depends on the number of measurements [27]. This study therefore includes shot noise in every primary analysis.

F. Fair comparison with classical reservoirs

Advantage claims depend on the strength of the classical baseline. A July 2026 survey concludes that current results do not establish a broad quantum advantage over well-matched classical reservoirs, and calls for matched baselines and complete resource accounting [31]. A companion study tuned classical baselines as carefully as the quantum models in noiseless simulations of up to 11 qubits [30]. Both QRC mechanisms tested, two-point correlator readout and feedback-driven QRC (cf. Ref. [41]), lost their advantage. On a hidden Markov-switching autoregressive prediction task, the tuned ESN reached a normalised RMSE of 0.720 ± 0.060 against 0.787 ± 0.059 for feedback QRC, over 10×10 seeds. The author limits the claim to these simulable sizes. Earlier advantage claims against ESNs [16, 29] and against a physically analogous classical system [13] should be read against this standard.

G. Research gap

The literature relates quantumness to absolute performance, not to the gap against a fair classical baseline. Palacios et al. [20] already compare mutual information, coherence and negativity against IPC, so the novelty here is narrower. Table I summarises what remains open and how each hypothesis addresses it.

III. RESEARCH QUESTION AND HYPOTHESES

Research question. In small spin-based quantum reservoirs under local decoherence, does the QMI of the input-driven reservoir state, computed without task targets, explain variation in the NARMA-10 performance gap to a matched, tuned ESN beyond what entanglement negativity and ℓ_1-norm coherence explain? Is that relationship the same under pure dephasing and amplitude damping?

Definitions. The performance gap Δ (Sec. IV F) is the ESN's NRMSE minus the quantum reservoir's NRMSE, so $\Delta > 0$ means the quantum reservoir is better. Diagnostics are time-averaged over the input-driven state after washout

Table 1: Open points in the literature and how this study addresses them.

Open point	Closest existing work	What this study adds	Hypothesis
Target variable	Diagnostics vs. short-term memory, parity or IPC [17, 18, 19, 20]	Predicts the gap Δ to a tuned, capacity-matched ESN	H1a
Channel	Bit flip and phase flip [20]; oscillator loss [37]	Amplitude damping vs. pure dephasing for the diagnostic–performance relation	H1b
Task	NARMA with memory capacity as indicator [26]	NARMA-10 with the three-diagnostic comparison	H1a, H1b
Transfer	Generality for local Markovian noise suggested, tested at $N = 5$ only [20]	Held-out channel, size ($N = 8$) and task	H1c
Measurement noise	Shown to reverse conclusions [20]	Shot noise in every primary analysis	All
Fair baseline	Advantage disappears under matched baselines [30, 31]	A diagnostic of when a positive gap exists	All

and never use the targets. The primary QMI is the multipartite total mutual information T; pairwise QMI is secondary (Sec. IV G). All performance figures include finite-shot noise. The hypotheses are summarised in Table II.

Rationale. QMI counts classical as well as quantum correlations. In Ref. [20], classical correlations carried most of the remaining capacity in the localised regime under strong phase-flip noise, while negativity behaved unlike the other measures. QMI is therefore most likely to add information in strongly dephased or localised regimes, and least likely in the ergodic regime where it tracks coherence (H1a). Amplitude damping is non-unital and produces processing capacity that unital channels do not [14]; it can improve performance where dephasing and depolarising noise degrade it [25, 26]. Against this, bit-flip and phase-flip data collapse onto one capacity–coherence curve [20], so a channel-independent relationship is plausible (H1b). The value of entanglement depends on input frequency under dissipation [18] and on the number of measurements [27]; a diagnostic that transfers across channels, sizes and tasks would be a stronger result than an in-sample correlation (H1c).

Interpretation of outcomes. H1a together with H1c would establish a transferable pre-task diagnostic. H1a without H1c would give a regime-specific diagnostic, with the regimes mapped. Equivalence on H0a would show that QMI is redundant with coherence and negativity, a clean negative result. A gap that is never positive under the fair baseline would be a fair-comparison result in the sense of Ref. [30].

IV. METHODS

This version of Section IV describes the pilot exactly as it was implemented and run (design v2, reduced-size sweep at $N = 4$), not the pre-registered confirmatory study. It replaces the design-v1 parameters of the original draft (washout $= 500$, $N \in \{4, 6\}$ main / $N = 8$ transfer, uncorrected $\alpha = 0.05$ margins, QMI not normalised per qubit) with the design actually exercised in `QMI_diagnostics`. The companion version of this document ("full study" Methods) describes the confirmatory sweep that has not yet been run. Pilot numbers below are diagnostic, not confirmatory: they are reported to show the pipeline works and to motivate the main sweep, not as the paper's results (see Sec. V, still to be written, and the Discussion).

A. Reservoir model

The reservoir is a network of N qubits with all-to-all transverse-field Ising couplings and on-site disorder, following Refs. [6, 11, 20]:

$$H = \sum_{i<j} J_{ij}\, \sigma_i^x \sigma_j^x + \frac{1}{2} \sum_{i=1}^{N} (h + D_i)\, \sigma_i^z, \tag{1}$$

with $J_{ij} \sim \mathscr{U}[-J_s/2, J_s/2]$ and $D_i \sim \mathscr{U}[-W, W]$. Energies are in units of $J_s = 1$ and times in units of $1/J_s$ ($\hbar = 1$). A *site* is one grid point (N, γ, h, W); the two channels at a site reuse the same Hamiltonian realisations (a paired design), and the pilot draws 4 realisations per site. Because the mean adjacent-gap ratio of a single $N = 4$ Hamiltonian is too noisy to classify a dynamical regime on its own, regime labels (ergodic / localised / other) are assigned per (h, W) point from a reference ensemble of 100 realisations at $N = 10$, not from the $N = 4$ spectra themselves; the per-configuration gap ratio is still recorded as a control.

Input injection. At step k a scalar input $s_k \in [0, 1]$ is written into qubit 1 by the reset-and-write map of Refs. [6, 20],

$$\rho_k = \mathscr{E}_{\Delta t}\big[\, |\psi_{s_k}\rangle\langle\psi_{s_k}| \otimes \mathrm{Tr}_1(\rho_{k-1}) \,\big], \tag{2}$$

$$|\psi_s\rangle = \sqrt{1-s}\,|0\rangle + \sqrt{s}\,|1\rangle, \tag{3}$$

where $\mathscr{E}_{\Delta t}$ is evolution for one input period Δt under the dynamics of Sec. IV B.

Table 2: Hypotheses, null hypotheses and tests. The margins (0.02 in R^2, 0.70 transfer ratio) are fixed in the pre-registration.

ID	Alternative hypothesis	Null hypothesis	Test
H1a	Adding QMI to a model of Δ containing negativity, coherence and controls raises cross-validated R^2 by at least 0.02	H0a: the gain is below 0.02	Nested models with cross-validation grouped by grid cell; redundancy claimed only if a TOST places the gain inside ±0.02
H1b	The QMI–Δ slope differs between pure dephasing and amplitude damping	H0b: no channel × QMI interaction	Interaction term, Holm-corrected across QMI variants; TOST on the standardised slope difference to claim channel independence
H1c	A model fitted in one setting keeps at least 70% of its in-setting R^2 on a held-out channel, on $N = 8$, and on a second task	H0c: the transfer ratio is below 0.70 in at least one held-out setting	Transfer ratio = held-out R^2 / in-setting R^2; reject H0c when the lower 95% bootstrap bound is ≥ 0.70

Table 3: Decoherence channels used in the pilot.

Channel	Jump operator	Unital	Role
Pure dephasing	$L_i = \frac{\sqrt{\gamma}}{2}\sigma_i^z$	Yes	Main comparison (H1b)
Amplitude damping	$L_i = \sqrt{\gamma}\sigma_i^-$	No	Main comparison (H1b)

B. Decoherence channels and simulation

Decoherence acts locally on every qubit through the Lindblad master equation

$$\dot{\rho} = -i[H,\rho] + \sum_{i=1}^{N}\left(L_i\rho L_i^\dagger - \tfrac{1}{2}\{L_i^\dagger L_i,\rho\}\right) \tag{4}$$

with the jump operators of Table III. Both pilot channels damp single-qubit coherences at rate $\gamma/2$ ($T_2 = 2/\gamma$), so the comparison is at equal coherence decay; amplitude damping additionally relaxes populations at rate γ ($T_1 = 1/\gamma$) toward $|0\rangle$.

States are dense $2^N \times 2^N$ density matrices. For $N = 4$, the only system size used in this pilot, each sub-interval applies the *exact* propagator $\exp(\mathscr{L}\tau)$ of the vectorised Liouvillian rather than a Trotter/Strang approximation, so no integrator error is present in the pilot results. (For $N \geq 5$, used only in the full study, a calibrated second-order Strang splitting is used instead; see the companion Methods version.)

C. Readout and finite sampling

Within each period, the expectation values $\langle\sigma_i^z\rangle$ and $\langle\sigma_i^z\sigma_j^z\rangle$ are recorded at $V = 10$ equally spaced times (time multiplexing), giving $D = 100$ features plus a bias at $N = 4$. Finite sampling is part of every primary analysis: each observable is replaced by its sampled value under the Gaussian limit of the multinomial distribution, with variance $(1-\langle O\rangle^2)/n_{\text{shots}}$ and the correlations of the state preserved across observables. The primary setting is $n_{\text{shots}} = 10^6$ (results tagged `s6` below); 10^5 and 10^4 are secondary. Measurement back-action is not modelled; an ensemble or restart protocol is assumed.

D. Tasks and capacity

The primary task is NARMA-10 [32], with $u_k \sim \mathscr{U}[0, 0.5]$ and

$$y_{k+1} = 0.3\,y_k + 0.05\,y_k\sum_{i=0}^{9} y_{k-i} + 1.5\,u_{k-9}u_k + 0.1. \tag{5}$$

The reservoir receives $s_k = 2u_k$. Each of the 5 shared input realisations has a washout of **2,000 steps** (fixed from a separate echo-state pilot, see Sec. IV J; this is the single largest change from the original design-v1 draft, which used a washout of only 500 steps), followed by 2,000 training, 500 validation and 1,000 test steps. Two further tasks are scored on the same reservoir trajectories as sensitivity/transfer checks within this pilot: NARMA-5 and NARMA-20 (tanh-bounded variant), and one-step Mackey–Glass prediction ($\tau = 17$).

E. Readout and classical baseline

All readouts are ridge regressions with an unpenalised bias; λ is chosen from 25 log-spaced values in $[10^{-10}, 10^2]$ by validation NRMSE and scored on the test set,

$$\text{NRMSE} = \sqrt{\langle(y_k - \hat{y}_k)^2\rangle / \text{Var}(y)}. \tag{6}$$

The classical baseline is a leaky echo state network [1, 3] with D nodes (matching the quantum readout dimension), tuned with a tree-structured Parzen estimator over leak rate, spectral radius, input scaling and bias scaling (100 trials per task), scored on the test set over 10 fresh weight seeds. The quantum reservoir receives no comparable tuning: only λ is fitted, which is conservative for the quantum side. An untuned ESN and a linear–quadratic regression on the last 10 inputs are reported as reference baselines.

F. Performance gap

For configuration c with readout dimension D_c,

$$\Delta_c = \overline{\text{NRMSE}}_{\text{ESN}}(D_c) - \overline{\text{NRMSE}}_{\text{QRC}}(c), \tag{7}$$

Table 4: Pilot design (design v2, reduced size). This is *not* the pre-registered sweep; compare Table IV of the companion "full study" Methods version.

Factor	Pilot sweep (`pilot2`)	Reproduction of [20]
N	4 only	5
Channel	Dephasing, amp. damping	Bit/phase flip
γ (units of J_s)	8 log-spaced, 10^{-3}–1	As in [20]
h	0.1, 1, 10	As in [20]
W	0.1, 5, 20	As in [20]
Realis. per site	4	As in [20]
Sites	72	—
Configurations	576	—
Outputs	NARMA-10/5/20, MG, diagnostics	IPC, coherence, negativity

with bars denoting means over the five input realisations (and the ten weight seeds for the ESN). $\Delta_c > 0$ means the quantum reservoir is better.

G. State diagnostics

Diagnostics are computed from the reservoir state alone and never see task targets, averaged over a 200-step post-washout window. Entropies use base-2 logarithms. The diagnostics are mean negativity $\bar{N}$, ℓ_1 coherence C_{ℓ_1} (normalised by $2^N - 1$), total mutual information T (the primary QMI, normalised **per qubit**, T/N — a second design-v2 change from the original draft, which left T un-normalised), and pairwise QMI $\bar{I}$ (secondary). Every model also includes controls: $\log\gamma$, h, W, gap ratio $\bar{r}$, channel, and purity in a sensitivity analysis.

H. Experimental design

The unit of analysis is a configuration: one Hamiltonian realisation under one channel at one site. Of the 576 pilot configurations, **78 failed the echo-state check** (below) and were excluded, leaving **498 configurations at 72 sites**. Channel comparisons use only the 44 of 72 sites where at least 80% of realisations pass under both channels (352 configurations, "common support").

I. Statistical analysis

All predictors are standardised within the fitting set. Cross-validation is 5-fold, grouped by site and repeated 10 times. Uncertainty comes from a cluster bootstrap over sites (2,000 resamples). Following design v2, the three confirmatory hypotheses are each tested at $\alpha = 0.05/3 \approx 0.0167$ (Bonferroni), so support intervals are 98.3% and equivalence intervals 96.7% — tighter than the uncorrected $\alpha = 0.05$ margins used in the original draft. H1a compares $M_0 : \Delta \sim \bar{N} + C_{\ell_1} + \text{controls}$ against $M_1 : M_0 + T$ by the cross-validated R^2 gain; H1b adds a channel$\times T$ interaction on common-support sites, with standard errors clustered by site and Holm correction across QMI variants; H1c tests whether a model fitted on one setting (e.g. one channel, or NARMA-10) keeps at least 70% of its explained variance on another (the other channel, or a second task), via a linear-recalibrated transfer ratio. The pilot does not test transfer across system size, since it was run at a single N.

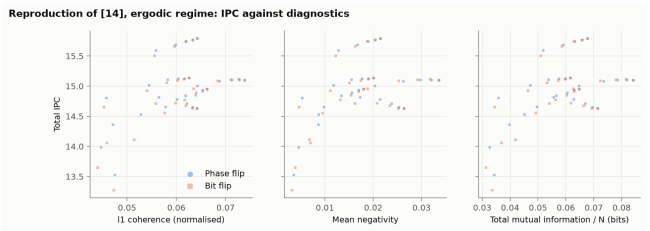


Figure 1: Reproduction of Ref. [20]: total IPC versus ℓ_1 coherence, mean negativity and total mutual information per qubit, for $N = 5$ in the ergodic regime under bit-flip and phase-flip noise (Sec. IV J). IPC rises monotonically with coherence and with QMI, while negativity saturates early and falls before IPC does — reproducing the qualitative pattern of Ref. [20] and validating the simulation pipeline used for the pilot (real data, generated by `scripts/reproduce_palacios.py`; axis label "[14]" in the figure is the source repository's internal numbering and corresponds to Ref. [20] in this paper).

J. Validation

Before the pilot sweep, the following checks were run and passed: unit tests on reference states (Bell pair, $|+\rangle^{\otimes N}$, product states) and channel Kraus/Choi properties; integrator accuracy (exact propagator used throughout at $N = 4$, so no splitting error); the echo-state property, checked per configuration (two random initial states driven by the same washout input must end within trace distance 10^{-6}; configurations failing this were counted and excluded, see Sec. IV H above); a qualitative reproduction of Ref. [20]'s IPC–coherence and IPC–negativity relations at $N = 5$ under bit-flip and phase-flip noise; and baseline sanity (the tuned ESN beats the untuned ESN and the linear–quadratic reference on every task).

K. Reproducibility

Every run records its configuration hash, seeds, code version and package versions. Raw results are stored as Parquet tables, and every figure and table reporting pilot numbers is produced by a script from those tables. Code: `https://github.com/sughraazim2008/QMI_diagnostics`.

V. RESULTS

[TODO: To be written after the main sweep; this is the paper's confirmatory Results section and must report the pre-registered sweep of Table IV in the "full study" Methods version, not pilot numbers. Suggested order:]

1. Validation: integrator accuracy, echo-state pass rate, and the reproduction of Ref. [20] (Fig. 1).
2. The gap Δ across channels, noise rates and regimes.
3. H1a: added value of QMI, pooled and by channel and regime.
4. H1b: channel dependence of the QMI–Δ slope.
5. H1c: transfer across channel, size and task.

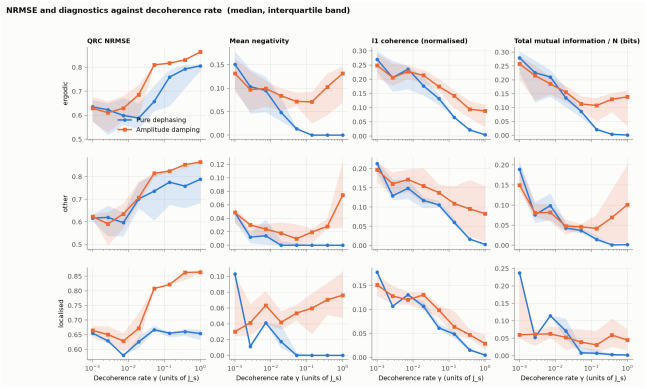


Figure 2: Pilot data. QRC NRMSE and the three diagnostics (mean negativity, ℓ_1 coherence, total mutual information per qubit) against decoherence rate γ, by dynamical regime and channel, median and interquartile band over the pilot's 4 Hamiltonian realisations per site. NRMSE rises with γ in all regimes (the quantum reservoir gets worse as decoherence strengthens), amplitude damping degrades performance somewhat faster than dephasing at strong γ, and all three diagnostics fall monotonically with γ — consistent with the pooled-redundancy result below (H1a).

6. Sensitivity: shot number, encoding, nonlinear models.

Pilot figures (not confirmatory). The four figures below are real output of the reduced-size ($N = 4$) pilot sweep described in Sec. IV, included here so the pipeline and plotting code can be checked. They are diagnostic, not the paper's results: the hypothesis tests they illustrate are run at pilot scale only and are superseded by the main sweep.

VI. DISCUSSION

[TODO: Interpret the results against Sec. II. Limitations to address: small system sizes ($N \leq 8$); the ensemble or restart readout without back-action; the asymmetric tuning budget (conservative for QRC); the choice of channel matching; and the dependence on one Hamiltonian family. The pilot figures above are evidence for planning the main sweep, not for this section's conclusions.]

VII. CONCLUSION

[TODO: Two or three sentences on the answer to the research question, once the main sweep (Table IV, companion Methods version) has been run.]

Data and Code Availability

Simulation and analysis code: `https://github.com/sughraazim2008/QMI_diagnostics`. [TODO: Zenodo DOI and pre-registration (OSF) link, once the main sweep is pre-registered.]

Acknowledgments

[TODO: Funding and acknowledgements.] The simulation pipeline was developed with AI pair-programming assistance (Claude, Anthropic); all design decisions, physics checks and reported numbers were reviewed by the author.

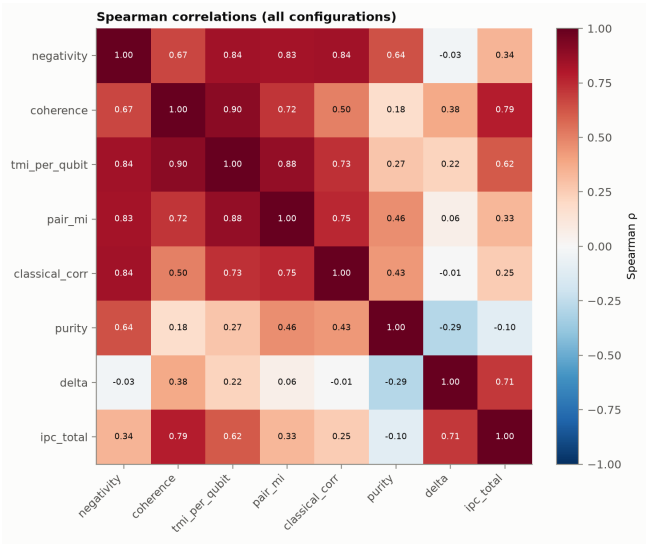


Figure 3: Pilot data. Spearman correlations among the three diagnostics, purity, the performance gap Δ and total IPC, over all 498 analysed pilot configurations. QMI (`tmi_per_qubit`) correlates strongly with both coherence ($\rho = 0.90$) and negativity ($\rho = 0.84$), which is the pilot-scale signature of the redundancy reported for H1a (Sec. IV I): the three diagnostics move together, so none adds much once the others are in the model. Note Δ is never positive in this pilot (Table in the companion Discussion), so its correlations with the diagnostics describe variation in how far behind the quantum reservoir falls, not whether it wins.

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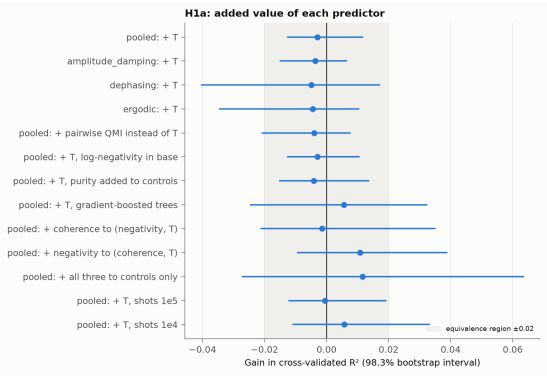


Figure 4: Pilot data. H1a sensitivity analysis: the gain in cross-validated R^2 from adding QMI to a model already containing negativity and coherence, pooled and under each robustness variant, with 98.3% cluster-bootstrap intervals and the pre-registered ± 0.02 equivalence region shaded. Every point estimate sits inside or very close to the equivalence region and every interval straddles zero: at pilot scale, QMI adds no cross-validated predictive value beyond negativity and coherence under any variant tested.

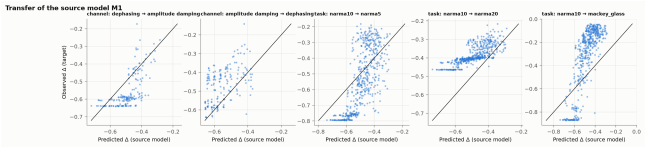


Figure 5: Pilot data. H1c transfer checks: predictions of the source model M_1 (QMI, negativity, coherence and controls) against the observed gap Δ in five held-out targets — both channel directions and three second tasks. The diagonal line is perfect transfer. By the pre-registered rule (transfer ratio $= r^2_{\text{target}}/R^2_{\text{cv,source}}$, lower 98.3% bootstrap bound ≥ 0.70), two of the five pilot-scale transfers already clear the bar: NARMA-10→NARMA-20 (ratio 0.95, lower bound 0.73) and NARMA-10→Mackey–Glass (ratio 1.07, lower bound 0.83). The other three do not: NARMA-10→NARMA-5 (ratio 0.81, lower bound 0.52) and both channel directions, dephasing→amplitude damping (ratio 0.79, lower bound 0.32) and the reverse (ratio 0.42, lower bound 0.13). Channel transfer is visibly the noisiest of the five at this sample size.

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